

ECE 3317

Homework #5

Assigned: Wednesday April 8

Due: Monday April 20

Note: In this homework set we are in the sinusoidal steady-state (phasor domain). Please use Smith charts for each problem, clearly showing your work.

1) From the Smith chart find Γ_L for the following Z_L^N :

(a) $0.6 - j0.4$; (b) 0 ; (c) ∞ ; (d) $3.5 + j2.0$

2) From the Smith chart, find the normalized input admittances corresponding to the following normalized input impedances:

(a) $Z_{in}^N = 0.7 - j0.3$; (b) $Z_{in}^N = 4 + j3$; (c) $Z_{in}^N = j\infty$

3) A transmission line is terminated in a normalized load impedance of $Z_L^N = 2.0 - j(1.5)$.

a) Indicate this position on the Smith chart with an “A”. Find the normalized load admittance and mark it with a “B”. What is the normalized load admittance?

b) Use the Smith chart to find the reflection coefficient at the load (both magnitude and phase).

c) Sketch $|V(z)|$, assuming $|V^+(z)| = 1$. Do this by using the Smith chart as a crank diagram. Indicate clearly on your diagram the values of V_{max} and V_{min} and their positions.

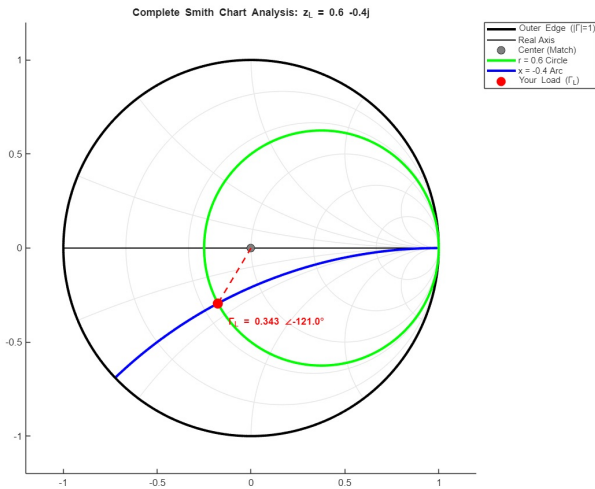
4) A $50\ [\Omega]$ transmission line (i.e., the characteristic impedance is $50\ [\Omega]$) has an unknown load impedance Z_L . The SWR is 1.8 and a voltage minimum occurs at $z = -0.15\lambda$. Find Z_L using the Smith chart.

1) From the Smith chart find Γ_L for the following Z_L^N :

- (a) $0.6 - j0.4$; (b) 0; (c) ∞ ; (d) $3.5 + j2.0$

(a) $0.6 - j0.4$;

Using the smith chart:



Handwritten calculations

$$Z_L^N = 0.6 - j0.4$$

$$Z_0 = 1$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = \frac{(0.6 - j0.4) - 1}{(0.6 - j0.4) + 1} = \frac{-0.4 - j0.4}{1.6 - j0.4} \cdot \frac{1.6 + j0.4}{1.6 + j0.4} =$$

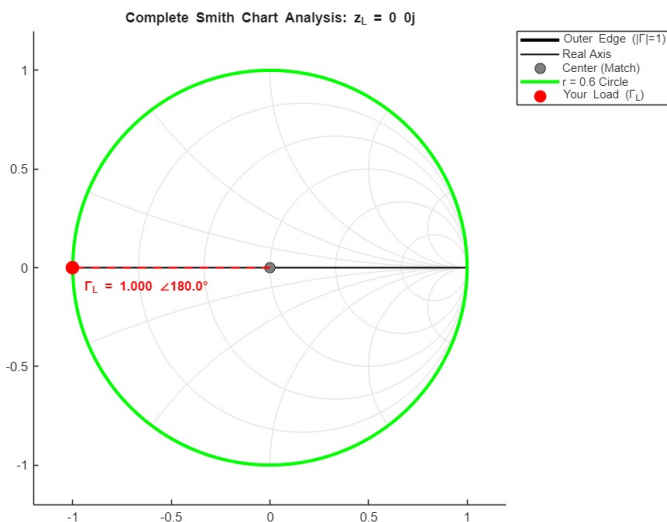
$$\frac{-0.64 - 0.16j - 0.64j + 0.16}{2.56 + 0.16} = \frac{-0.48 - 0.8j}{2.72} = -0.1765 - 0.2941j$$

$$|\Gamma_L| = \sqrt{0.1765^2 + 0.2941^2} = 0.3430$$

$$\angle \Gamma_L = \tan^{-1} \left(\frac{-0.2941}{-0.1765} \right) = 59^\circ - 180^\circ = -121^\circ$$

(b) 0;

Using the smith chart:



Handwritten calculations

$$Z_L^N = 0$$

$$Z_0 = 1$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

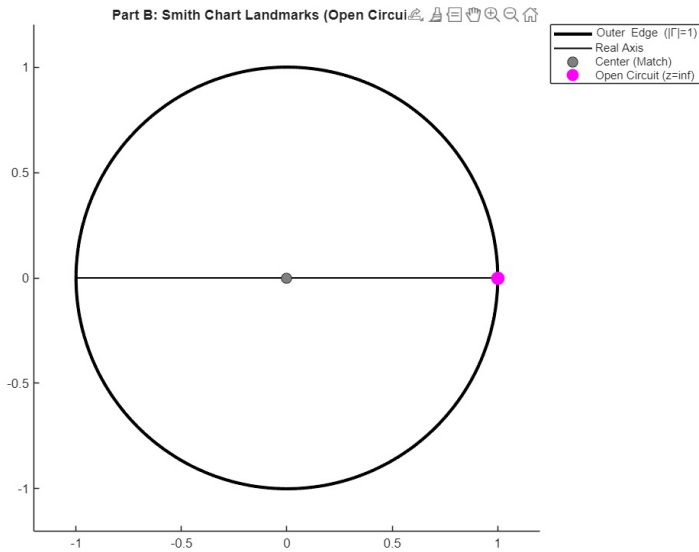
$$\Gamma_L = \frac{0 - 1}{0 + 1} = -1$$

$$|\Gamma_L| = 1$$

$$\angle \Gamma_L = \tan^{-1} = 0^\circ + 180^\circ = 180^\circ$$

(c) ∞ ;

Using the smith chart:



Handwritten calculations

$$Z_L^N = \infty \quad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Z_0 = 1$$

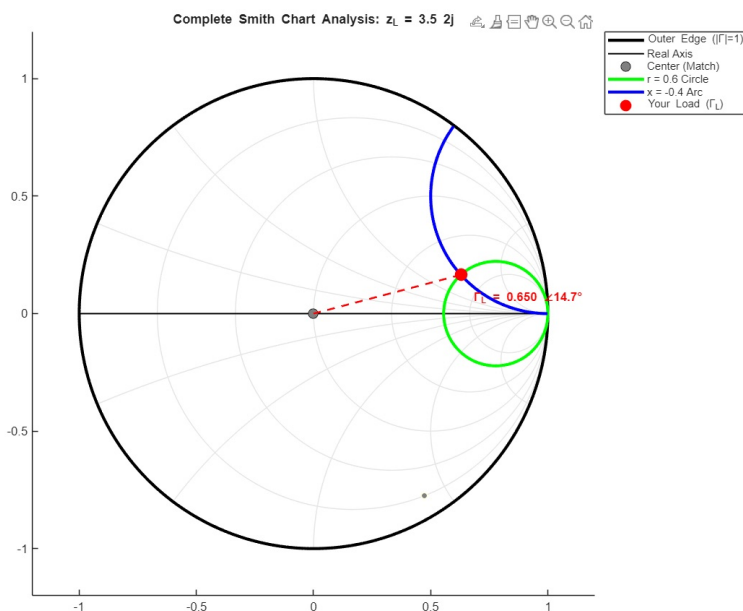
$$\Gamma_L = \frac{\infty - 1}{\infty + 1} = 1$$

$$|\Gamma_L| = 1$$

$$\angle \Gamma_L = \tan^{-1} = 0^\circ$$

(d) $3.5 + j2.0$

Using the smith chart:



Handwritten calculations

$$Z_L^N = 3.5 + j2.0 \quad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Z_0 = 1$$

$$\Gamma_L = \frac{3.5 + j2.0 - 1}{3.5 + j2.0 + 1} = \frac{2.5 + j2.0}{4.5 + j2.0} \cdot \frac{4.5 - j2.0}{4.5 - j2.0}$$

$$\frac{11.25 - 5.0j + 9.0j + 4}{20.25 + 4} = \frac{15.25 + 4.0j}{24.25} = 0.6289 + j0.1649$$

$$|\Gamma_L| = \sqrt{0.6289^2 + 0.1649^2} = 0.6501$$

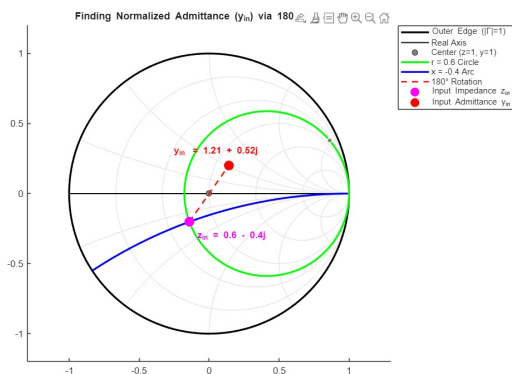
$$\angle \Gamma_L = \tan^{-1}\left(\frac{0.1649}{0.6289}\right) = 14.6924^\circ$$

2) From the Smith chart, find the normalized input admittances corresponding to the following normalized input impedances:

- (a) $Z_{in}^N = 0.7 - j0.3$; (b) $Z_{in}^N = 4 + j3$; (c) $Z_{in}^N = j\infty$

(a) $Z_{in}^N = 0.7 - j0.3$;

Using the smith chart:



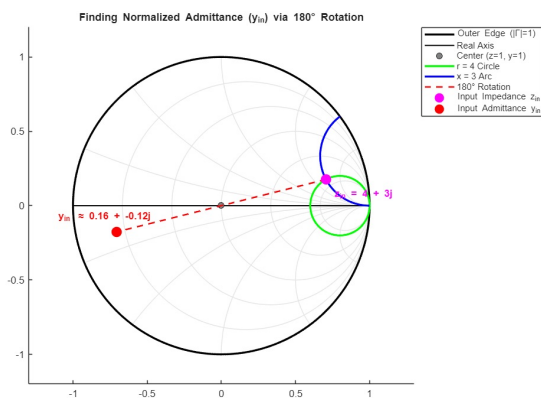
Handwritten calculations:

$$a.) \frac{1}{0.7 - 0.3j} = \frac{1}{0.7 - 0.3j} \cdot \frac{0.7 + 0.3j}{0.7 + 0.3j}$$

$$= \frac{0.7 + 0.3j}{0.49 + 0.09} = \frac{0.7 + 0.3j}{0.58} = 1.2069 + 0.5172j$$

(b) $Z_{in}^N = 4 + j3$;

Using the smith chart:



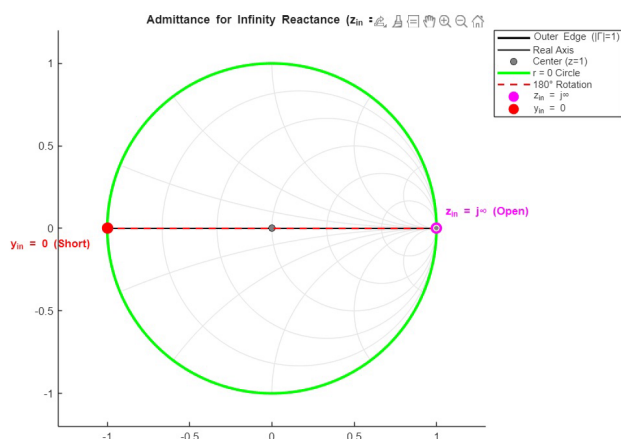
Handwritten calculations:

$$\frac{1}{4 + 3j} = \frac{1}{4 + 3j} \cdot \frac{4 - 3j}{4 - 3j}$$

$$= \frac{4 - 3j}{16 + 9} = \frac{4 - 3j}{25} = 0.16 - 0.12j$$

(c) $Z_{in}^N = j\infty$

Using the smith chart:



Handwritten calculations:

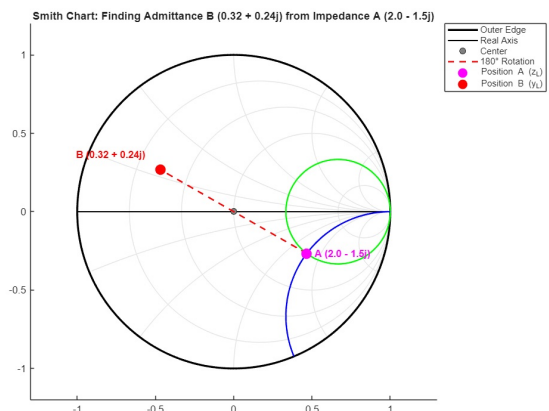
$$\frac{1}{\infty j} = 0$$

3) A transmission line is terminated in a normalized load impedance of $Z_L^N = 2.0 - j(1.5)$.

- Indicate this position on the Smith chart with an "A". Find the normalized load admittance and mark it with a "B". What is the normalized load admittance?
- Use the Smith chart to find the reflection coefficient at the load (both magnitude and phase).
- Sketch $|V(z)|$, assuming $|V^+(z)| = 1$. Do this by using the Smith chart as a crank diagram. Indicate clearly on your diagram the values of V_{max} and V_{min} and their positions.

a)

Using the smith chart:



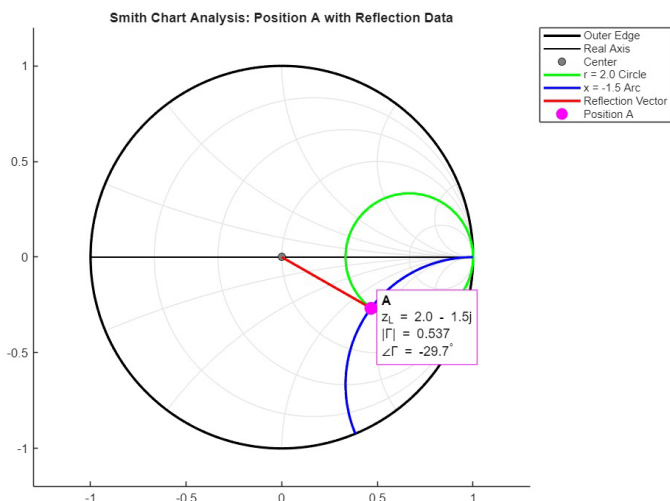
Handwritten calculations:

$$\frac{1}{2.0 - 1.5j} = \frac{1}{2.0 - 1.5j} \cdot \frac{2.0 + 1.5j}{2.0 + 1.5j}$$

$$= \frac{2.0 + 1.5j}{4.0 + 2.25} = \frac{2.0 + 1.5j}{6.25} = 0.32 + 0.24j$$

b)

Using the smith chart:



Handwritten calculations:

$$Z_L^N = 2.0 - 1.5j$$

$$\Gamma_L = \frac{Z_L^N - 1}{Z_L^N + 1} = \frac{(2.0 - 1.5j) - 1}{(2.0 - 1.5j) + 1} = \frac{1.0 - 1.5j}{3.0 - 1.5j} \cdot \frac{3.0 + 1.5j}{3.0 + 1.5j}$$

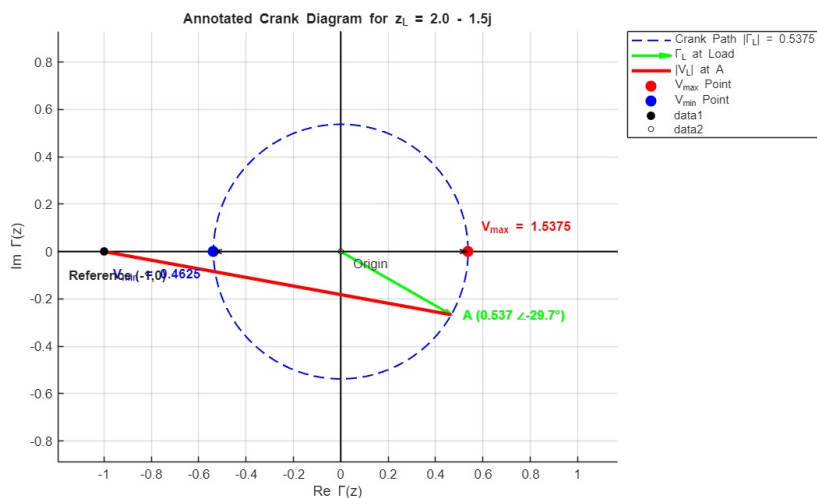
$$= \frac{3.0 + 1.5j - 4.5j + 2.25}{9.0 + 2.25} = \frac{5.25 - 3.0j}{11.25} = 0.4667 - 0.2667j$$

$$|\Gamma_L| = \sqrt{0.4667^2 + 0.2667^2} = 0.5375$$

$$\angle \Gamma_L = \tan^{-1}\left(\frac{-0.2667}{0.4667}\right) = -29.75^\circ$$

c.)

Using the smith chart:



Handwritten calculations:

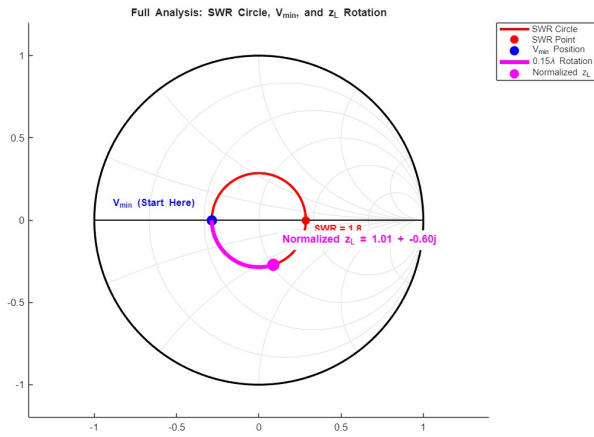
$$V_{max} = 1 + |\Gamma_L| = 1 + 0.5375 = 1.5375 [V]$$

$$V_{min} = 1 - |\Gamma_L| = 1 - 0.5375 = 0.4625 [V]$$

Handwritten calculations:

- 4) A $50 \text{ } [\Omega]$ transmission line (i.e., the characteristic impedance is $50 \text{ } [\Omega]$) has an unknown load impedance Z_L . The SWR is 1.8 and a voltage minimum occurs at $z = -0.15\lambda$. Find Z_L using the Smith chart.

Using the smith chart:



$$Z_L = Z_L^N \cdot 50 [\Omega] = (1.01 - 0.6j) (50) = [50.5 - 30j] [\Omega]$$

$$|\Gamma_L| = \frac{1.8 - 1}{1.8 + 1} = \frac{0.8}{2.8} = \frac{2}{7} \approx 0.2857$$

$$Z_L = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right), \quad Z_0 = 1$$

$$|\Gamma_L| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

$$\text{From } z : \theta_L = 2\beta z - \pi$$

$$\beta = \frac{2\pi}{\lambda}$$

$$\theta_{\Gamma_L} = 2 \left(\frac{2\pi}{\lambda} \right) (-0.15\lambda) - \pi = -0.6\pi - \pi = -1.6\pi$$

$$\angle \theta_{\Gamma_L} = -1.6\pi \Rightarrow -1.6\pi \cdot \frac{180}{\pi} = -288$$

$$0.2857 \cos(288^\circ) \approx 0.0883$$

$$0.2857 \sin(288^\circ) \approx -0.2717$$

$$\Gamma_L = 0.0883 - 0.2717j$$

$$Z_L^N = \frac{1 + (0.0883 - 0.2717j)}{1 - (0.0883 - 0.2717j)} = 1.0148 - 0.6004j$$

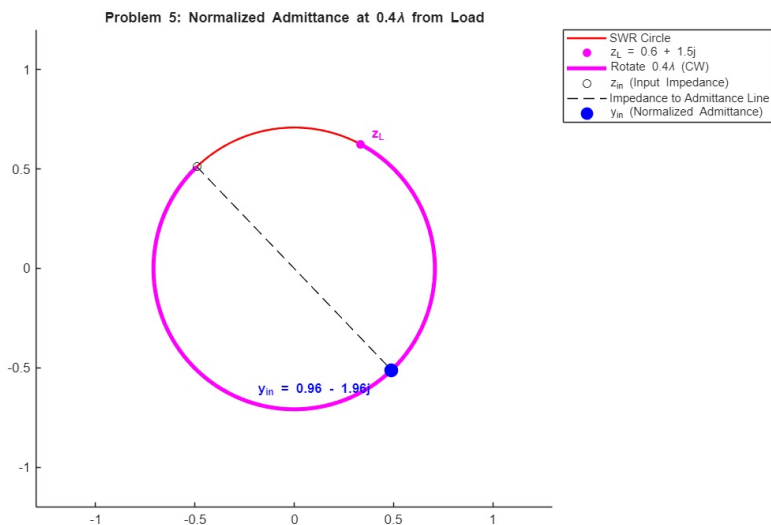
$$Z_L = 50.7384 - 30.0215j$$

- 5) A $50\ [\Omega]$ transmission line has a load impedance of $Z_L = 30 + j(75)\ [\Omega]$. Find the normalized load admittance at a distance of 0.4λ from the load, using the Smith chart.
- 6) A transmission line has a normalized load impedance of $Z_L^N = 0.5 - j(0.2)$. The transmission line has a characteristic impedance of $50\ [\Omega]$. Find the stub length l_s (in terms of λ) of a short-circuited stub to be placed at the load, and the characteristic impedance Z_{0T} of a quarter-wave transmission line connected to the load, which will provide a quarter-wave transformer matching system for the line. Assume that the characteristic impedance of the stub line is also $50\ [\Omega]$. Use either exact equation or Smith chart to find the stub length.
- 7) A transmission line has a normalized load impedance of $Z_L^N = 0.5 - j(0.2)$. Find the stub position d and the stub length l_s (in terms of λ) in order to provide a single stub matching system for the line. Use the Smith chart. (Show both possible solutions.) Assume that both the main line and the stub line have the same characteristic impedance.

- 5) A $50\ [\Omega]$ transmission line has a load impedance of $Z_L = 30 + j(75)\ [\Omega]$. Find the normalized load admittance at a distance of 0.4λ from the load, using the Smith chart.

$$Z_L^N = \frac{30 + j75}{50} = 0.6 + j1.5j$$

Using the smith chart:



Handwritten calculations:

$$Z_L^N = \frac{30 + j75}{50} = 0.6 + j1.5j$$

$$d = 0.4\lambda$$

$$Z_{in}(d) = Z_0 \frac{Z_L + jZ_0 \tan(\beta d)}{Z_0 + jZ_L \tan(\beta d)}$$

$$Z_0 = 50, \quad \beta = \frac{2\pi}{\lambda}$$

$$in(0.4\lambda) = 1 \cdot \left(\frac{(0.6 + j1.5j) + j \tan(0.8\pi)}{1 + j(0.6 + j1.5j) \tan(0.8\pi)} \right)$$

$$\tan(0.8\pi) = -0.7265 \text{ rad}$$

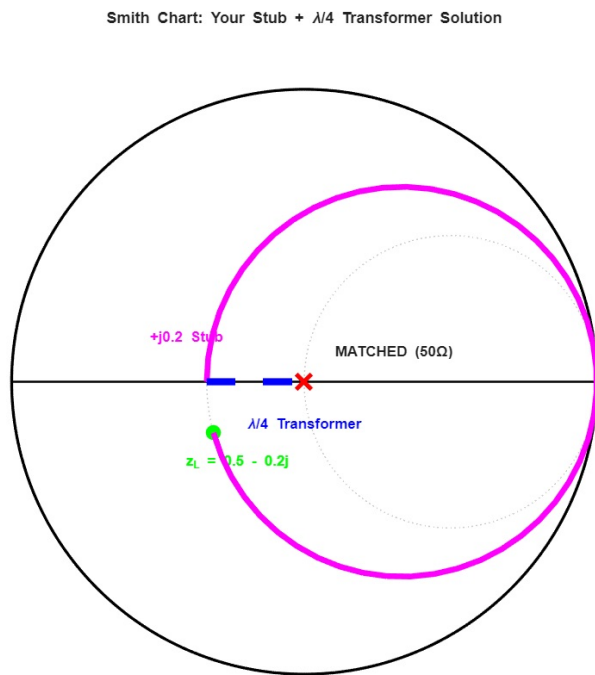
$$Z_{in} = \frac{0.6 + j1.5j - 0.7265j}{1 - 0.7265j(0.6 + j1.5j)} = \frac{0.6 + 0.7735j}{1 + (1.0898 - 0.4359j)}$$

$$Z_{in} \approx 0.2011 + 0.4121j$$

$$Y_{in} \approx \frac{1}{Z_{in}} \approx 0.9566 - j1.9597j$$

- 6) A transmission line has a normalized load impedance of $Z_L^N = 0.5 - j(0.2)$. The transmission line has a characteristic impedance of $50 [\Omega]$. Find the stub length l_s (in terms of λ) of a short-circuited stub to be placed at the load, and the characteristic impedance Z_{0T} of a quarter-wave transmission line connected to the load, which will provide a quarter-wave transformer matching system for the line. Assume that the characteristic impedance of the stub line is also $50 [\Omega]$. Use either exact equation or Smith chart to find the stub length.

Using the smith chart:



Handwritten calculations:

To cancel $(-0.2j)$, you need to be able to find and stub of $0.2j$.

$$Z_{in}^{stub} = jZ_{0s} \tan(\beta_s l_s)$$

$$\beta = \frac{2\pi}{\lambda}$$

$$0.2j = j \tan\left(\left(\frac{2\pi}{\lambda}\right) l_s\right)$$

$$\tan^{-1}(0.2) = \frac{2\pi}{\lambda} l_s$$

$$\tan^{-1}(0.2) = 0.1974$$

$$0.1974 = \frac{2\pi}{\lambda} l_s$$

$$l_s = 0.0314 \lambda$$

$$Z_{0T} = \sqrt{Z_0 R_L}$$

$$R_L = 0.5 \cdot 50$$

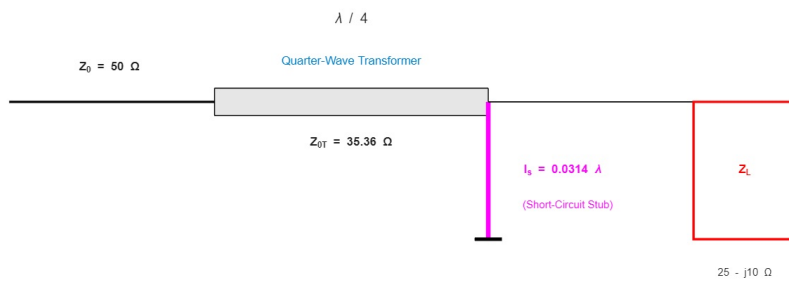
$$R_L = 25 [\Omega]$$

$$Z_0 = 50 [\Omega]$$

$$Z_{0T} = \sqrt{50 \cdot 25}$$

$$Z_{0T} = 35.36 [\Omega]$$

Circuit Layout: Short-Circuited Stub + Quarter-Wave Transformer



Step 1: Stub cancels $-j0.2$ reactive part.

Step 2: $\lambda/4$ line transforms $R_L = 25 \Omega$ to 50Ω .

- 7) A transmission line has a normalized load impedance of $Z_L^N = 0.5 - j(0.2)$. Find the stub position d and the stub length l_s (in terms of λ) in order to provide a single stub matching system for the line. Use the Smith chart. (Show both possible solutions.) Assume that both the main line and the stub line have the same characteristic impedance.

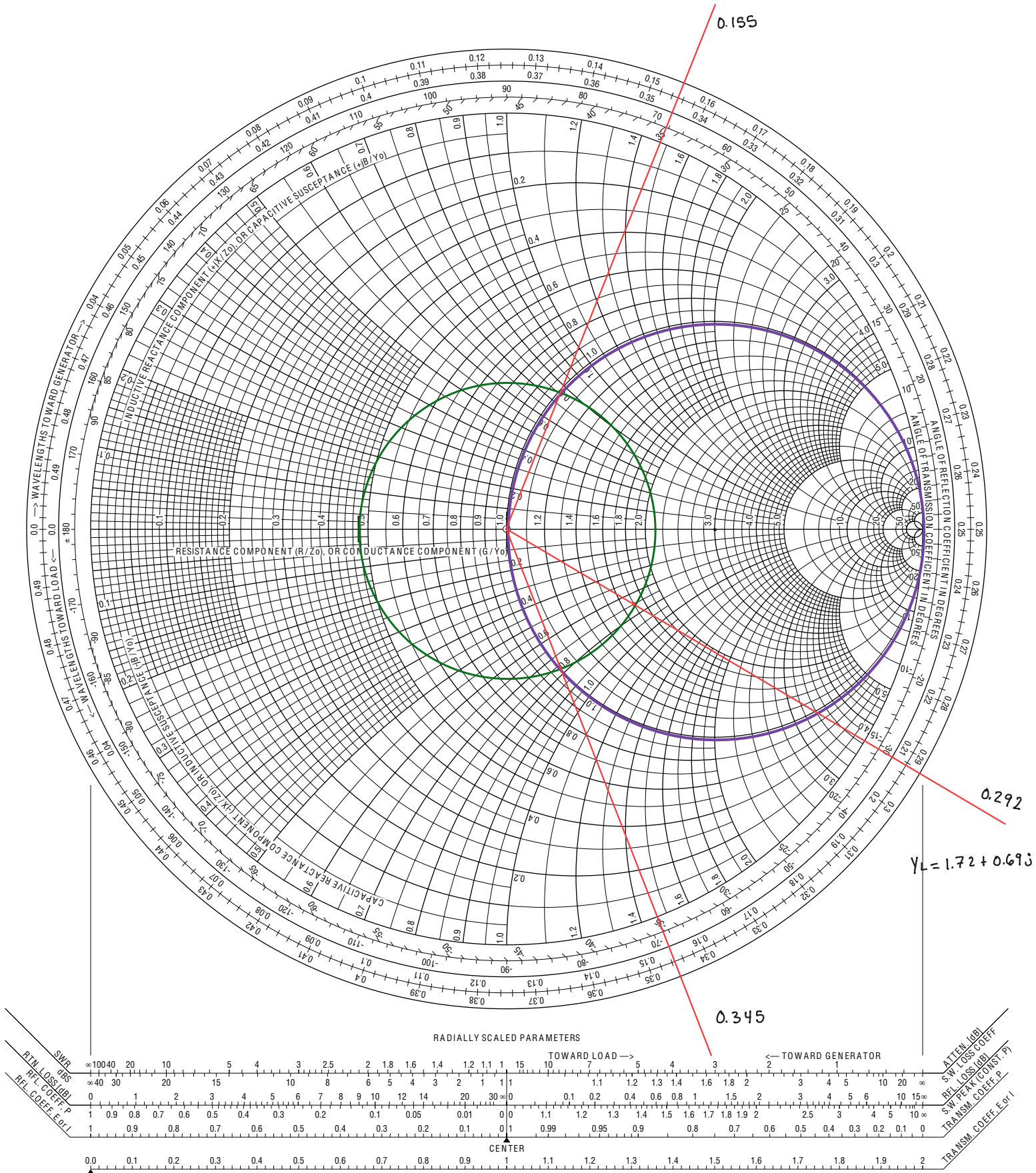
Using the smith chart:

Handwritten calculations:

$$\begin{aligned} \gamma_L &= 0.460\lambda \\ d_1 &= 0.097\lambda \quad y(d_1) = 1 - j0.76 \\ d_2 &= 0.404\lambda \quad y(d_2) = 1 + j0.76 \\ l_{s1} &= 0.353\lambda \\ l_{s2} &= 0.147\lambda \\ \gamma_L &= 1.72 + 0.69j \end{aligned}$$

Smith Chart

.363
.093



NAME	TITLE	DWG. NO.
SMITH CHART FORM ZY-01-N	Microwave Circuit Design - EE523 - Fall 2000	DATE

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

